

A Systematic Review of Machine Learning Approaches for Predicting Sepsis in Patients

Disha Patel, Mir Patel, Samarth Patel, Shubh Patel

1 Introduction

Sepsis is a life-threatening syndrome of organ dysfunction caused by a dysregulated immune response to infection [1]. It imposes a massive global burden, with an estimated 48.9 million cases and 11 million sepsis-related deaths annually (about 20% of all global deaths in 2017) [1]. Clinical outcomes hinge on timely intervention – each hour of delayed treatment increases mortality by approximately 7–8% [1]. Early identification is therefore paramount to improve survival in sepsis.

Despite this urgency, early sepsis detection in practice remains challenging. Traditional clinical tools like the Systemic Inflammatory Response Syndrome (SIRS) criteria, Sequential Organ Failure Assessment (SOFA), and quick SOFA (qSOFA) have notable limitations in sensitivity and specificity [2]. These score-based frameworks often rely on coarse vital-sign thresholds and clinician assessment, leading to inconsistent recognition of sepsis in its initial stages [2]. In fact, such methods tend to flag sepsis only after significant organ dysfunction has occurred, missing the critical window for early intervention [6]. For example, qSOFA – though simple – is now discouraged as a lone screening tool due to its weak predictive power for incipient sepsis [14].

In recent years (2019–2025), machine learning (ML) has emerged as a powerful approach to anticipate sepsis before full-blown clinical manifestation [2]. By leveraging electronic health record (EHR) data – vital signs, laboratory trends, demographics, histories, and clinician notes – ML models can detect subtle patterns indicative of evolving sepsis risk [15]. Many outperform traditional scoring systems in predictive accuracy [7].

Studies show ML-based early warning systems can detect sepsis earlier and more reliably than SIRS/qSOFA, yielding higher AUROC values [4]. The use of open critical care datasets like MIMIC-III, MIMIC-IV, eICU, HiRID, and AmsterdamUMCdb has accelerated development [3]. Using these, ensemble and deep learning models have achieved AUROCs of 0.83–0.93, offering prediction horizons up to 48 hours before onset [5].

Explainable AI (XAI) techniques like SHAP and LIME improve interpretability and clinician trust [17]. Hospitals deploying interpretable ML sepsis models have reported outcome

improvements such as reduced ICU stays and mortality [18]. This review therefore examines ML-based sepsis prediction from 2019–2025, highlighting datasets, model types, interpretability, and clinical integration challenges.

2 Background and Related Work

2.1 Clinical Background of Sepsis

Sepsis is defined by the Sepsis-3 consensus as a life-threatening organ dysfunction caused by an unregulated host response to infection [8]. This revised definition represents a substantial progression from previous criteria, prioritizing organ dysfunction over inflammatory response [9]. Sepsis constitutes a major global public health burden, with 48.9 million cases reported in 2017, resulting in 11 million deaths, accounting for approximately 20% of all deaths. The geographic distribution of sepsis reveals significant disparities across regions. Sub-Saharan Africa, Oceania, and South Asia experience the highest burden, with some areas reporting more than 1,500 cases per 100,000 people annually [11]. Children under 5 years old account for nearly half of all sepsis cases, highlighting their particular vulnerability to this condition [12]. The rapid progression of sepsis presents a critical clinical challenge, as the condition can prove fatal within 12 hours without appropriate treatment [13]. The transition from mild symptoms to septic shock and multiple organ failure can occur rapidly. This swift progression underscores the critical importance of early detection and timely intervention in sepsis management. Each hour of delayed treatment significantly increases mortality risk, making rapid identification essential for improving patient outcomes [13].

2.2 Traditional Sepsis Detection Methods and Limitations

Current sepsis detection methods rely primarily on clinical scores and laboratory markers, each with distinct limitations. The Systemic Inflammatory Response Syndrome (SIRS) criteria, which requires two or more abnormalities in temperature, heart rate, respiratory rate, or white blood cell count, demonstrated high sensitivity but poor specificity, leading to frequent false positives [13]. The Sequential Organ Failure Assessment (SOFA) score provides strong prognostic accuracy by evaluating organ dysfunction through laboratory and clinical parameters, but its requirement for extensive laboratory testing limits bedside applicability [14]. The quick SOFA (qSOFA) score, utilizing only three bedside variables—systolic blood pressure ≤ 100 mmHg, respiratory rate ≥ 22 , and Glasgow Coma Scale ≤ 15 —offers improved specificity but suffers from low sensitivity, frequently missing early sepsis cases [14,15]. The National Early Warning Score (NEWS) provides composite vital sign scoring for detecting general patient deterioration, but lacks infection specificity and demonstrates variable accuracy across populations [15]. Serum lactate levels above 2 mmol/L serve as markers of tissue hypoperfu-

sion, but elevated lactate is non-specific and can occur in various other conditions [16]. Blood culture, while representing the diagnostic gold standard for microbiological confirmation, suffers from slow turnaround times and low sensitivity [15]. These traditional methods share three fundamental limitations: inadequate sensitivity and specificity, late-stage detection after organ dysfunction has progressed, and significant performance variability across different populations and clinical contexts. These constraints demonstrate the urgent need for machine learning-based approaches capable of capturing broader signals for earlier and more reliable sepsis identification at the point of care.

Table 1: Comparison of Traditional Sepsis Detection Methods

Tool/Method	Basis of Detection	Strengths	Limitations	Ref.
SIRS	≥ 2 abnormalities in temp, HR, RR, WBC	Simple, sensitive	Poor specificity; false positives	[12]
SOFA	Organ dysfunction (lab + clinical)	Prognostic accuracy	Requires labs; not bedside-friendly	[13]
qSOFA	SBP ≤ 100 , RR ≥ 22 , GCS ≤ 15	Bedside simplicity	Low sensitivity; misses early sepsis	[13, 14]
NEWS	Composite vital score	Detects deterioration	Not infection-specific	[14]
Lactate	> 2 mmol/L	Hypoperfusion marker	Non-specific	[12]
Blood culture	Microbiological confirmation	Diagnostic gold standard	Slow, low sensitivity	[16]

2.3 Publicly Available Sepsis Datasets

The availability of large-scale, publicly accessible datasets has been instrumental in advancing machine learning research for sepsis prediction. The MIMIC-III database, released in 2016, contains data from approximately 53,000 ICU stays and has become the foundational dataset for sepsis prediction research [18]. This comprehensive dataset includes vital signs, laboratory values, medications, and clinical notes, enabling development of various machine learning models. The more recent MIMIC-IV database, released in 2023, expands upon its predecessor with approximately 70,000 ICU stays and enhanced laboratory and electronic health record data. The eICU Collaborative Research Database, published in 2019, represents a multicenter dataset encompassing over 200,000 ICU stays from diverse healthcare facilities. This dataset enables external validation studies and assessment of model generalizability across different clinical settings. The HiRID database, introduced in 2021, contains 33,000 ICU episodes with high-frequency vital sign recordings, making it particularly valuable for real-time early detection research. The AmsterdamUMCdb database, also released in 2021, provides data from 23,000 ICU stays in an international context. This dataset supports investigation of model performance across different healthcare systems and patient populations. Dataset diversity remains crucial for developing robust and generalizable sepsis prediction models capable of performing reliably across varied clinical environments. The variability in sepsis definitions, data collection protocols, and patient demographics across these datasets presents both challenges and opportunities for developing more robust prediction algorithms.

Table 2: Overview of Publicly Available Sepsis Datasets

Dataset	Year	Size	Key Features	Reference
MIMIC-III	2016	~53k ICU stays	Vitals, labs, notes	[16]
MIMIC-IV	2023	~70k ICU stays	Expanded EHR data	[3]
eICU	2019	~200k stays	Multicenter dataset	[3]
HiRID	2021	33k ICU episodes	High-frequency vitals	[3]
AmsterdamUMCdb	2021	23k ICU stays	International dataset	[3]

2.4 Machine Learning Approaches in Sepsis Prediction

Machine learning techniques for sepsis prediction have evolved significantly from traditional statistical methods to sophisticated deep learning architectures. Logistic regression served as the baseline approach, with systems like InSight demonstrating superior performance compared to conventional scoring systems such as qSOFA and SIRS [14]. Random Forest and Support Vector Machine algorithms showed improved performance over simple statistical methods in handling complex clinical datasets [16]. Sequential Minimal Optimization variants demonstrated reasonable accuracy rates in classification tasks with computational efficiency advantages. Long Short-Term Memory (LSTM) networks emerged as powerful tools for modeling temporal patterns in sepsis progression using electronic health records [17]. The SepsisLab system effectively captured time-series dependencies in vital signs and laboratory values through LSTM architectures [17]. Gated Recurrent Unit (GRU) models provided computational efficiency while maintaining temporal modeling capabilities with comparable performance [17]. Deep neural networks demonstrated superior feature extraction capabilities compared to traditional feature engineering approaches in complex clinical scenarios. Ensemble approaches combining multiple algorithms showed enhanced predictive performance through model diversity and reduced overfitting. XGBoost implementations demonstrated state-of-the-art performance in several sepsis prediction benchmarks with robust handling of missing values [13]. The integration of structured electronic health record data with unstructured clinical notes represents a promising multimodal direction for comprehensive patient assessment [13]. Feature engineering remained crucial across all machine learning approaches, with careful selection of clinical variables significantly impacting model performance. Area Under the Receiver Operating Characteristic curve emerged as the primary evaluation metric across sepsis prediction studies [14]. The InSight system achieved AUROC values of 0.880 at sepsis onset time with superior performance compared to traditional scoring systems [14]. Prediction horizons varied significantly across studies, with most focusing on 4-hour to 48-hour advance warning capabilities for clinical intervention [17]. The SepsisLab system demonstrated effective prediction up to 4 hours before sepsis onset with maintained accuracy across diverse patient populations [17]. Performance improvement programs incorporating machine learning algorithms showed significant increases in compliance with sepsis bundles and mortality reduction [10]. Early ad-

ministration of antibiotics and intravenous fluids remained crucial components of sepsis treatment protocols regardless of prediction methodology [10]. Active sensing algorithms were developed to recommend optimal laboratory tests for reducing prediction uncertainty in clinical decision-making processes [17]. Cross-validation techniques ensured robust model evaluation across diverse patient populations and clinical settings throughout sepsis prediction studies. (Table 3).

Table 3: Machine Learning Models for Sepsis Prediction

Study	Year	Dataset	Model	Horizon	AUROC
Desautels et al.	2016	MIMIC-III	Logistic Regression (InSight)	At onset	0.880
Alanazi et al.	2023	ICU datasets	Random Forest, SVM	4–48h	Variable
Yin et al.	2024	MIMIC-IV	LSTM (SepsisLab)	4h	High accuracy
Various	2019–24	Multiple	XGBoost	4–48h	State-of-art

2.5 Explainable Artificial Intelligence in Sepsis Prediction

Explainable Artificial Intelligence (XAI) plays a vital role in sepsis prediction by building clinical trust and supporting safe deployment of machine learning models in real-world health-care settings. Many advanced machine learning algorithms function as “black boxes,” making it difficult for clinicians to understand how predictions are made, which often limits adoption [19,20]. XAI addresses this challenge by offering clarity and actionable insights, allowing clinicians to make faster and more informed decisions during critical triage and sepsis management [19,20]. Among the most widely discussed XAI techniques are SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME). SHAP, grounded in strong theoretical foundations, distributes contribution values fairly across features to explain how individual predictions differ from the overall model baseline [19,21,20]. It is particularly effective for tree-based models and helps quantify both the magnitude and direction of each variable’s influence on the outcome. LIME offers quick, case-specific interpretations by breaking down the prediction probability and showing the weight of each feature [22,20]. This allows clinicians to understand exactly why a particular patient was flagged, making the method both versatile and easy to apply in practice. The impact of XAI in sepsis prediction extends beyond interpretability alone. Tools like SHAP help pinpoint the most influential predictors, highlighting key clinical parameters such as vital signs (systolic blood pressure, heart rate, oxygen saturation, respiratory rate, temperature) and laboratory findings (albumin, lactate, SOFA score, Glasgow Coma Scale) [19,20]. This guides clinicians to focus on the most relevant patient data, improving both diagnostic accuracy and timeliness of interventions. XAI fosters greater trust and acceptance among clinicians by making models more transparent, with successful implementations integrating clinician feedback through iterative Plan-Do-Study-Act (PDSA) cycles [22]. These iterative improvements led to refinements like optimized alert thresholds

and a “snooze” option—changes that were “well accepted by clinicians” [22]. By improving alert specificity, XAI helps reduce “alarm fatigue,” a common issue in critical care, thereby boosting clinicians’ confidence in the system [20]. Clinician training programs are essential to ensure that model outputs are correctly understood and effectively acted upon [23]. XAI supports the pathway to regulatory approval and broader clinical adoption, with growing demand for external validation across diverse datasets and adherence to best practices in reporting and implementing AI in healthcare [14,15]. Transparent and interpretable models provide the accountability, evidence, and reliability required for safer, more generalizable, and clinically valuable sepsis prediction tools. The integration of XAI techniques represents a critical step toward bridging the gap between advanced machine learning capabilities and practical clinical implementation.

2.6 Existing Systematic Reviews and Their Gaps

Several systematic reviews have attempted to synthesize machine learning approaches for sepsis prediction over the past decade without comprehensive coverage. Most existing surveys concentrated on algorithmic comparisons rather than practical deployment considerations in clinical environments and real-world implementation challenges [16]. Geographic bias toward North American and European healthcare settings limited the applicability of review conclusions across diverse patient populations [16]. Early systematic reviews demonstrated significant gaps in dataset diversity coverage, with heavy emphasis on MIMIC-III database analyses. Existing reviews inadequately addressed the interpretability and explainability aspects crucial for clinical acceptance of machine learning models in practice. The black-box nature of complex algorithms posed significant barriers to clinical implementation that previous reviews failed to adequately discuss [19,20]. Uncertainty quantification methods received minimal attention despite their critical importance in high-stakes clinical decisions and patient safety considerations [1,2]. Missing data handling strategies, despite their critical importance in clinical datasets, were inadequately covered in previous systematic review frameworks. Transformer-based architectures and attention mechanisms for sepsis prediction received minimal coverage in existing systematic reviews [1]. Multimodal approaches combining structured data with unstructured clinical notes through natural language processing remained underrepresented in literature syntheses [16]. Graph neural networks and their application to clinical relationship modeling were absent from previous review syntheses. Federated learning approaches for privacy-preserving sepsis prediction across multiple healthcare institutions received insufficient attention. Real-world deployment challenges, including computational requirements and system integration issues, remained underexplored in previous systematic review frameworks [10]. Cost-effectiveness analyses and economic impact assessments were rarely included in existing systematic review methodologies. The integration of clinical workflow considerations and human-AI collaboration aspects were notably absent from previous reviews [1,2]. Re-

inforcement learning applications for dynamic treatment recommendations remained largely unexplored in previous systematic syntheses. This comprehensive review addresses existing gaps by providing systematic coverage of transformer-based architectures and multimodal fusion approaches for sepsis prediction. We emphasize explainable AI techniques and uncertainty quantification methods crucial for clinical decision-making processes and patient safety [19,20]. Our review synthesizes findings across diverse healthcare systems and geographic regions to enhance generalizability of conclusions and practical applicability. The integration of clinical workflow analysis and economic impact assessment distinguishes our review from previous syntheses in the literature.

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